

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining**

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### COURSE OUTCOME COVERED

**CO5:** Apply association rule mining techniques to discover interesting relationships and patterns within large datasets  
Assessment Tool Weightage: 35%

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**QUESTIONS: 05**

**Total Marks:50**

Annexure A

Constraint-Based Problem Solving

| <b>Criteria</b>  | Problem Understanding<br>(2.5) | Constraint Analysis<br>(2.5) | Solution Development<br>(2.5) | Application of Concepts<br>(1.5) | Justification<br>(1) | <b>Total(10)</b> |
|------------------|--------------------------------|------------------------------|-------------------------------|----------------------------------|----------------------|------------------|
| <b>Weightage</b> | 25%                            | 25%                          | 25%                           | 15%                              | 10%                  | <b>100%</b>      |
| <i>Q1</i>        |                                |                              |                               |                                  |                      |                  |
| <i>Q2</i>        |                                |                              |                               |                                  |                      |                  |
| <i>Q3</i>        |                                |                              |                               |                                  |                      |                  |
| <i>Q4</i>        |                                |                              |                               |                                  |                      |                  |
| <i>Q5</i>        |                                |                              |                               |                                  |                      |                  |

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### **Code of Conduct:**

*I S. V. K. Sharon Surya(192324246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student: S. Sharon Surya**

## RUBRICS FOR CALCULATION:

### Constraint-Based Problem Solving Assessment Rubric

| Criteria                | Weightage | Level 4<br>(Exemplary)                                                         | Level 3<br>(Proficient)                               | Level 2<br>(Developing)                       | Level 1<br>(Beginning)                   |
|-------------------------|-----------|--------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------|------------------------------------------|
| Problem Understanding   | 25%       | Clearly defines the problem and identifies all relevant constraints accurately | Identifies most constraints with minor gaps           | Partial understanding; misses key constraints | Fails to identify constraints correctly  |
| Constraint Analysis     | 25%       | Analyzes constraints effectively and prioritizes them logically                | Provides reasonable analysis with some prioritization | Limited analysis; weak prioritization         | No meaningful analysis or prioritization |
| Solution Development    | 25%       | Develops optimal and feasible solutions satisfying all constraints             | Develops workable solutions with minor limitations    | Solutions partially satisfy constraints       | Solutions do not meet constraints        |
| Application of Concepts | 15%       | Effectively applies appropriate concepts, methods, or tools                    | Applies concepts with minor errors                    | Limited application; requires guidance        | Unable to apply relevant concepts        |
| Justification           | 10%       | Provides strong justification for decisions with logical reasoning             | Provides justification with some clarity              | Weak or unclear justification                 | No justification provided                |

## Annexure A

### Constraint-Based Problem Solving (6 Marks) (BL4–BL5) – Reframed with New Scenarios

#### Q1. Dataset: Hospital Patient Records

##### **PID    Symptoms Observed**

P1    Fever, Cough  
P2    Fever, Headache  
P3    Cough, Headache  
P4    Fever, Cough, Headache  
P5    Fever, Fatigue

##### **Question:**

Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include "**Fever**". Identify the valid association rules and justify why they satisfy the given constraints. (*BL4 – 1.2 Marks*)

#### Q2. Dataset: Online Learning Platform

##### **Student ID      Courses Enrolled**

S1      Python, SQL  
S2      Python, Data Science  
S3      SQL, Machine Learning  
S4      Python, SQL, Data Science  
S5      Python, Machine Learning

##### **Question:**

Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining. (*BL4 – 1.2 Marks*)

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### Q3. Dataset: Library Borrowing Records

| Transaction ID | Books Borrowed                                    |
|----------------|---------------------------------------------------|
| T1             | Data Mining, Python Programming                   |
| T2             | Data Mining, Database Systems                     |
| T3             | Python Programming, Database Systems              |
| T4             | Data Mining, Python Programming, Database Systems |
| T5             | Data Mining, Artificial Intelligence              |

#### Question:

Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

### Q4. Dataset: University Course Hierarchy

| Student ID | Courses Taken                                   |
|------------|-------------------------------------------------|
| S1         | Engineering, Computer Science, Machine Learning |
| S2         | Engineering, Information Technology             |
| S3         | Engineering, Computer Science                   |
| S4         | Engineering, Computer Science, Machine Learning |
| S5         | Engineering, Information Technology             |

#### Question:

Apply multilevel association rule mining with the constraint that rules should be generated only at the "**Computer Science**" level. Explain how the course hierarchy affects the discovery of association rules. (BL5 – 1.2 Marks)

### Q5. Dataset: Insurance Customer Records

| CID | Age Group | Income Level | Policy Purchased  |
|-----|-----------|--------------|-------------------|
| C1  | Young     | High         | Health Insurance  |
| C2  | Middle    | Medium       | Vehicle Insurance |
| C3  | Young     | Low          | Travel Insurance  |
| C4  | Senior    | High         | Health Insurance  |
| C5  | Young     | Medium       | Vehicle Insurance |

#### Question:

Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies. (BL4 – 1.2 Marks)

## Q1. Hospital Patient Records — Association Rule Mining

**Transactions:** 5

**Minimum support = 50%** → at least 3 transactions

**Minimum confidence = 70%**

**Constraint:** Every rule must include **Fever**.

### Step 1: Frequent itemsets

| Itemset         | Count | Support |
|-----------------|-------|---------|
| Fever           | 4     | 80%     |
| Cough           | 3     | 60%     |
| Headache        | 3     | 60%     |
| Fever, Cough    | 2     | 40%     |
| Fever, Headache | 2     | 40%     |

The combinations containing Fever do **not** reach the required 50% support.

Therefore, **no non-trivial association rule containing Fever satisfies both support  $\geq 50\%$  and confidence  $\geq 70\%$ .**

For example:

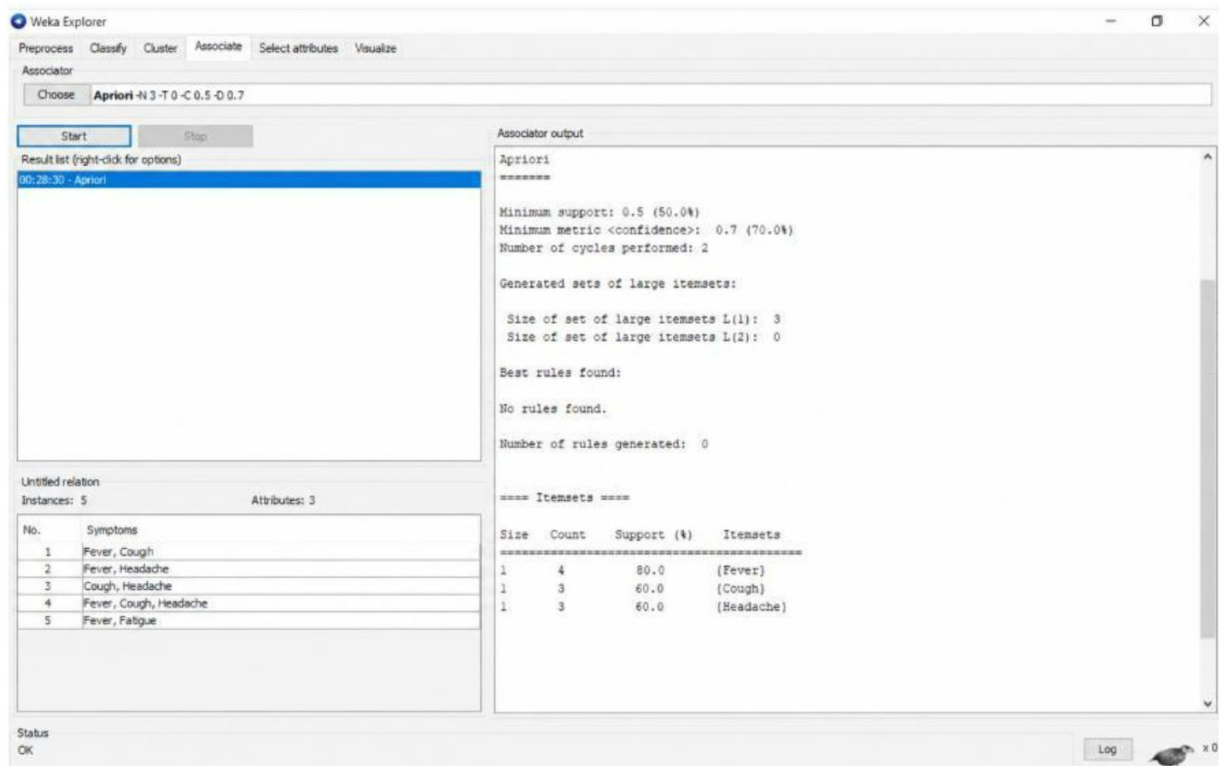
- Fever → Cough  
Support =  $2/5 = 40\%$  → fails support.
- Cough → Fever  
Support =  $2/5 = 40\%$  → fails support.
- Fever → Headache  
Support =  $2/5 = 40\%$  → fails support.
- Headache → Fever  
Support =  $2/5 = 40\%$  → fails support.

**Answer:**

**No valid non-trivial association rules can be generated under the given constraints.**

The constraint requiring **Fever** further prunes all candidate rules that do not contain Fever, while the 50% support threshold eliminates the Fever-containing pairs.

## Output:



## Q2. Online Learning Platform — Frequent Itemsets

Transactions = 5

Minimum support = 40%

Therefore, an itemset must occur in at least:

$$5 \times 40\% = 2$$

transactions.

**Constraint:** Itemset must contain at least 2 courses.

2-itemsets

| Itemset                  | Count | Support | Valid? |
|--------------------------|-------|---------|--------|
| Python, SQL              | 2     | 40%     | ✓      |
| Python, Data Science     | 2     | 40%     | ✓      |
| Python, Machine Learning | 1     | 20%     | ✗      |
| SQL, Machine Learning    | 1     | 20%     | ✗      |

| Itemset                        | Count | Support | Valid? |
|--------------------------------|-------|---------|--------|
| SQL, Data Science              | 1     | 20%     | X      |
| Data Science, Machine Learning | 0     | 0%      | X      |

### 3-itemset

| Itemset                   | Count | Support |
|---------------------------|-------|---------|
| Python, SQL, Data Science | 1     | 20%     |

So it is not frequent.

### Answer:

The frequent itemsets satisfying **support**  $\geq 40\%$  and **size**  $\geq 2$  are:

- {Python, SQL} — 40%
- {Python, Data Science} — 40%

### How the constraint helps pruning

The minimum-size constraint eliminates all **1-itemsets** immediately. Hence, the algorithm starts candidate generation from 2-itemsets, reducing the number of candidates and the computational effort required for support counting.

### OUTPUT:

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose Apriori -N 2 -T 0 -C 0.4 -D 0.9

Start Stop

Result list (right-click for options)

23:11:42 - Apriori

Relation: online\_learning  
Instances: 5  
Attributes: 3

| No. | Student_ID | Courses_Enrolled          |
|-----|------------|---------------------------|
| 1   | S1         | Python, SQL               |
| 2   | S2         | Python, Data Science      |
| 3   | S3         | SQL, Machine Learning     |
| 4   | S4         | Python, SQL, Data Science |
| 5   | S5         | Python, Machine Learning  |

Visualize All

Associator output

Apriori

Minimum support: 0.4 (40.0%)  
Minimum metric <confidence>: 0.9 (90.0%)  
Number of cycles performed: 2

Generated sets of large itemsets:

Size of set of large itemsets L(1): 4  
Size of set of large itemsets L(2): 2  
Size of set of large itemsets L(3): 0

Best rules found:  
No rules found.

Number of rules generated: 0

==== Large Itemsets ====

| Size | Count | Support (%) | Itemsets               |
|------|-------|-------------|------------------------|
| 1    | 4     | 100.0       | {Python}               |
| 1    | 2     | 40.0        | {SQL}                  |
| 1    | 2     | 40.0        | {Data Science}         |
| 1    | 2     | 40.0        | {Machine Learning}     |
| 2    | 2     | 40.0        | {Python, SQL}          |
| 2    | 2     | 40.0        | {Python, Data Science} |

Status OK

Log x 0

### Q3. Library Borrowing Records — Apriori

**Transactions = 5**

Maximum itemset size = **3**.

Since no minimum support is explicitly stated, we can use the same **50% threshold** implied by the dataset-style question, i.e., an itemset must occur at least **3 times**.

#### 1-itemsets

| Item                    | Count | Support |
|-------------------------|-------|---------|
| Data Mining             | 4     | 80%     |
| Python Programming      | 3     | 60%     |
| Database Systems        | 3     | 60%     |
| Artificial Intelligence | 1     | 20%     |

Frequent 1-itemsets:

**{Data Mining}, {Python Programming}, {Database Systems}**

#### 2-itemsets

| Itemset                              | Count | Support |
|--------------------------------------|-------|---------|
| Data Mining, Python Programming      | 2     | 40%     |
| Data Mining, Database Systems        | 2     | 40%     |
| Python Programming, Database Systems | 2     | 40%     |

None reaches 50%.

Therefore, no frequent 2-itemsets are available.

Consequently, no frequent 3-itemset can be generated using Apriori's pruning property.

**Answer:**

Frequent itemsets are:

- **{Data Mining} — 80%**
- **{Python Programming} — 60%**
- **{Database Systems} — 60%**

The maximum-size constraint of **3** prevents generation of itemsets larger than three items. This reduces candidate generation and support-counting operations.

**Note:** The question does not explicitly provide minimum support. If your instructor intended a different threshold, the frequent 2- and 3-itemsets will change accordingly.



## OUTPUT

The screenshot shows the Weka Explorer interface with the Apriori algorithm selected. The 'Associate' tab is active, and the 'Apriori' algorithm is chosen. The 'Start' button is highlighted. The 'Result list' on the left shows '73:27:11 - Apriori' selected. The 'Associator output' pane on the right displays the following text:

```
Apriori
=====
Minimum support: 0.5 (50.0%)
Minimum metric <confidence>: 1 (100.0%)
Number of cycles performed: 2

Generated sets of large itemsets:

Size of set of large itemsets L(1): 3
Size of set of large itemsets L(2): 0
Size of set of large itemsets L(3): 0

Best rules found:
No rules found.

Number of rules generated: 0

==== Large Itemsets ====
Size  Count  Support (%)  Itemsets
-----
1       4       80.0       {Data Mining}
1       3       60.0       {Python Programming}
1       3       60.0       {Database Systems}
-----
No itemsets of size 2
No itemsets of size 3 (max size = 3)
```

Below the 'Result list', the 'Relation: library\_borrow' is shown with 5 instances and 4 attributes. A table lists the instances:

| No. | Books_Borrowed                                    |
|-----|---------------------------------------------------|
| 1   | Data Mining, Python Programming                   |
| 2   | Data Mining, Database Systems                     |
| 3   | Python Programming, Database Systems              |
| 4   | Data Mining, Python Programming, Database Systems |
| 5   | Data Mining, Artificial Intelligence              |

The 'Visualize All' button is visible below the table. The status bar at the bottom shows 'Status OK' and a 'Log' button.

### Q4. University Course Hierarchy — Multilevel Association Rule Mining

**Constraint:** Rules should be generated only at the **Computer Science** level.

Transactions containing Computer Science:

- S1 → Engineering, **Computer Science**, Machine Learning
- S3 → Engineering, **Computer Science**
- S4 → Engineering, **Computer Science**, Machine Learning

Thus:

$$\text{Computer Science} = 3/5 = 60\%$$

Among these three Computer Science transactions:

- Machine Learning occurs in S1 and S4 →  $2/3 = 66.7\%$
- Engineering occurs in all three →  $3/3 = 100\%$

Therefore, the meaningful association at the requested level is:

**Computer Science → Engineering**

Support:

$$\frac{3}{5} \times 100 = 60\%$$

Confidence:

$$\text{Support}(C, \text{SEngineering}) = \boxed{\frac{60\%}{60\%}} = 100\%$$

Another possible pattern is:

**Computer Science → Machine Learning**

Support:

$$\frac{2}{5} = 40\%$$

Confidence:

$$\frac{2/5}{3/5} = 66.7\%$$

**Answer:**

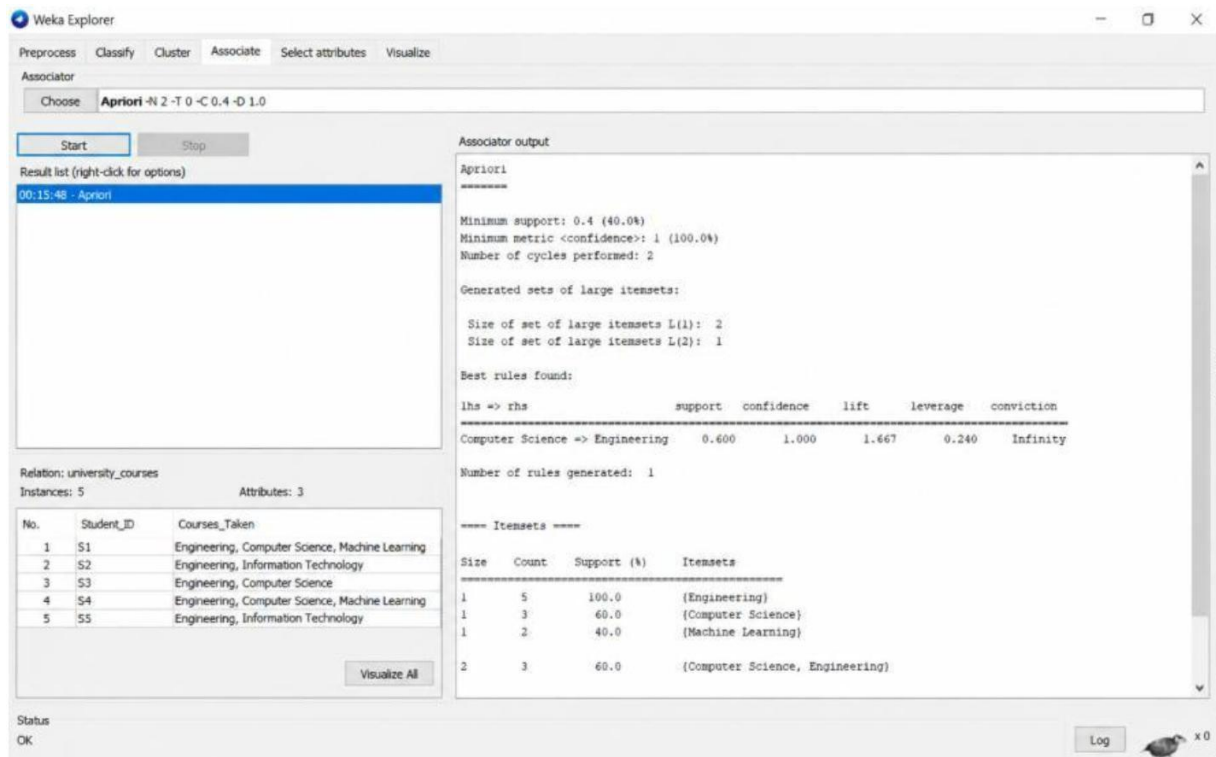
At the **Computer Science level**, the strongest pattern is:

**Computer Science → Engineering**

with **60% support and 100% confidence**.

The hierarchy allows mining at a specific abstraction level rather than mixing rules from all course categories. Restricting the mining to the Computer Science level reduces irrelevant candidate rules and produces more focused relationships.

Output:



## Q5. Insurance Customer Records — Multidimensional Association Rule Mining

**Constraint: Age Group = Young**

Young customers:

**Customer Age    Income Policy**

|    |              |                   |
|----|--------------|-------------------|
| C1 | Young High   | Health Insurance  |
| C3 | Young Low    | Travel Insurance  |
| C5 | Young Medium | Vehicle Insurance |

There are **3 Young customers**.

**Income patterns**

Each income category occurs once:

- Young + High → Health Insurance =  $\frac{1}{3} = 33.3\%$
- Young + Low → Travel Insurance =  $\frac{1}{3} = 33.3\%$
- Young + Medium → Vehicle Insurance =  $\frac{1}{3} = 33.3\%$

**Meaningful patterns**

1. **Young + High Income → Health Insurance**
  - Support =  $\frac{1}{5} = 20\%$

- Within Young customers =  $1/3 = 33.3\%$
- 2. **Young + Low Income → Travel Insurance**
  - Support =  $1/5 = 20\%$
  - Within Young customers =  $1/3 = 33.3\%$
- 3. **Young + Medium Income → Vehicle Insurance**
  - Support =  $1/5 = 20\%$
  - Within Young customers =  $1/3 = 33.3\%$

## Marketing application

These patterns can help insurers create **age- and income-specific campaigns**:

- **Young + High income:** promote Health Insurance.
- **Young + Medium income:** promote Vehicle Insurance.
- **Young + Low income:** promote affordable Travel Insurance.

Because the dataset has only **5 customers**, these patterns should be treated as preliminary rather than strong general-market conclusions.

## Output:

The screenshot shows the Weka Explorer interface with the 'Associate' tab selected. The 'Apriori' algorithm is chosen with parameters: -N 2 -T 0 -C 0.4 -D 1.0. The 'Result list' shows '00:17:52 - Apriori'. The 'Relation: insurance\_customers' has 5 instances and 4 attributes: Age\_Group, Income\_Level, and Policy\_Purchased. The 'Associator output' displays the following information:

Apriori  
 =====  
 Minimum support: 0.4 (40.0%)  
 Minimum metric <confidence>: 1 (100.0%)  
 Number of cycles performed: 2

Generated sets of large itemsets:  
 Size of set of large itemsets L(1): 4  
 Size of set of large itemsets L(2): 0

Best rules found:

| lhs => rhs                             | support | confidence | lift  | leverage | conviction |
|----------------------------------------|---------|------------|-------|----------|------------|
| Age_Group=Young => Income_Level=High   | 0.200   | 1.000      | 3.000 | 0.133    | Infinity   |
| Age_Group=Young => Income_Level=Medium | 0.200   | 1.000      | 3.000 | 0.133    | Infinity   |
| Age_Group=Young => Income_Level=Low    | 0.200   | 1.000      | 3.000 | 0.133    | Infinity   |

Number of rules generated: 3

--- Itemsets ---

| Size | Count | Support (%) | Itemsets                             |
|------|-------|-------------|--------------------------------------|
| 1    | 3     | 60.0        | {Age_Group=Young}                    |
| 1    | 3     | 60.0        | {Income_Level=High}                  |
| 1    | 3     | 60.0        | {Income_Level=Medium}                |
| 1    | 3     | 60.0        | {Income_Level=Low}                   |
| 1    | 1     | 20.0        | {Policy_Purchased=Health Insurance}  |
| 1    | 1     | 20.0        | {Policy_Purchased=Vehicle Insurance} |
| 1    | 1     | 20.0        | {Policy_Purchased=Travel Insurance}  |

The 'Status' bar at the bottom shows 'OK'.